

KARTIK MAHESH BAMBLE

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SUMMARY

Full-stack developer with 3 years of experience building scalable Java Spring Boot and serverless applications with RESTful APIs and modern JavaScript frameworks. Experienced implementing automation and observability on AWS using Lambda, OpenSearch, and CI/CD pipelines to improve reliability and reduce latency. Proven track record delivering a distributed search migration that cut latency by 40%, seeking to leverage full stack Java expertise in a growth-focused development team.

EDUCATION

Syracuse University | *Masters, Computer Science*

Aug 2024 - May 2026

University Of Mumbai | *Bachelor of Engineering, Computer Science*

Jun 2018 - May 2022

TECHNICAL SKILLS

- **Programming:** Python, Java, C#, Go, SQL, Shell, JavaScript, Vue.js, Node.js, GraphQL, RESTful APIs, Angular, React, TypeScript, HTML5, CSS
- **ML/Frameworks:** PyTorch, Transformers, FastAPI, Django, Flask, Spring Boot, RAG, LangChain, Agentic AI, LLMs, Prompt Engineering
- **Data and Distributed Systems:** MongoDB, PostgreSQL, MySQL, Neo4j, DuckDB, Redis, Splunk, Kafka, Spark, Snowflake, Delta Lake, Airflow, RabbitMQ
- **Tools:** AWS, Azure, Kubernetes, Docker, Git, AWS Glue, Athena, Databricks, Elasticsearch, Splunk, Redis, Linux, Jenkins, Ansible, Cursor, Claude, SOLR, Maven, Gradle, Bitbucket, Bamboo
- **Methodologies & Practices:** DevOps Practices, Automated Testing, Agile Software Development, Continuous Integration, Object-oriented Design Patterns

WORK EXPERIENCE

Commonwealth of Massachusetts | *Software Developer Intern*

Jun 2025 - Aug 2025

- Modernized a distributed enterprise search platform by migrating legacy IIS services to AWS Lambda and REST APIs with OpenSearch, reducing search latency by ~40%
- Built event-driven data ingestion pipelines using AWS Lambda and SQS, processing 1M+ records across 5 integrated systems for near-real-time indexing
- Implemented structured logging and CloudWatch metrics to enhance observability, reducing production incident MTTR by ~20%
- Participated in Agile scrum ceremonies and peer code reviews in DevOps under agile software development principles, contributing to reduced post-release defects and improved system reliability.

Tata Consultancy Services | *Software Engineer*

Aug 2023 - Jun 2024

- Spearheaded org-wide observability initiative using OpenTelemetry and Prometheus for real-time metrics collection, leveraging NLP-based log parsing for anomaly detection across 100+ services, reducing troubleshooting time by 80%
- Architected Kafka self-serve internal developer platform using Spring Boot and Python, following object-oriented design patterns to ensure maintainability and scalability, enabling engineering teams to independently manage topics, consumer groups, and configurations, eliminating JIRA-driven admin bottleneck and reducing provisioning time by 90%.
- Engineered scalable automation orchestration platform using Python, Kafka, and Redis, replacing proprietary licensed platforms and reducing operational costs by 80%
- Resolved critical security audit finding by automating Kubernetes secrets rotation using HashiCorp Vault Agent as sidecar, enabling automatic credential refresh and pod updates across all services, preventing 150K+ annual security incidents
- Implemented continuous integration pipelines using Bitbucket and Bamboo to automate builds with Maven, reducing integration issues by 75% and accelerating deployment cycles.

Junior Software Engineer

Jul 2022 - Aug 2023

- Built ETL data pipeline automating terabytes of compliance data transfer to S3 Glacier Deep Archive, saving \$1.2M annually in storage costs
- Delivered pipeline orchestration using AWS Lambda and Step Functions for compliance case management, reducing manual workload by 95%
- Authored automated test suite using pytest and moto for AWS Lambda and Step Functions compliance pipelines, achieving 90%+ coverage and preventing production regression

DY Patil University | *Full Stack Developer*

Aug 2020 - Mar 2021

- Launched full-stack student transcript management portal using Django and PostgreSQL with a responsive UI built using Angular, TypeScript, HTML5, and CSS, enabling self-service transcript generation and request workflows for 5,000+ students and alumni.
- Engineered ML-based course recommendation engine using collaborative filtering and scikit-learn, automating lead capture via RESTful APIs into Zoho CRM through FastAPI and AWS Lambda webhooks, improving enrollment conversion from 18% to 35%.

PROJECTS

Provenance

- Built a multi-agent LangGraph research assistant using Python and FastAPI RESTful APIs to search, filter, and synthesize findings from academic papers, achieving 0.81 RAGAS faithfulness with a citation-verification agent grounding every claim

RepoMind MCP

- Developed a codebase intelligence tool using GraphRAG on Neo4j to map GitHub repositories into function-class-dependency graphs; containerized the MCP server with Docker and automated CI/CD pipelines using GitHub Actions to expose RESTful query APIs for AI clients (Claude Desktop, Cursor)

MedFineTune

- Fine-tuned Mistral-7B on clinical QA datasets (PubMedQA + MedQA) using QLoRA (4-bit, r=64), improving MedQA accuracy from 45% to 57% over base model; served production inference via vLLM OpenAI-compatible API

Dotboard

- Real-time collaborative pixel-art board with a Go WebSocket server broadcasting live updates to concurrent users, with per-user rate limiting and persisted board state

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate, AWS Certified AI Practitioner, Confluent Data Streaming Engineer, Hashicorp Terraform Associate